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A000112(014)

B.Tech., PQ (First Semester) Examination 2020

(AIC Scheme)

(AEI, Bio Tech, Chem., Civil, CSE, Elect., EEE, EI, Et & T, IT, Mech., Mining, Metallurgy, Mechatronics, Production, Automobile Agriculture, Plastic Br.)

MATHEMATICS - I

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *All questions are to be attempted. Part (a) is compulsory in each unit and carries 4 marks. Attempt any two parts from (b), (c) and (d) of each unit which carry 8 marks each.*

Unit-I

1. (a) (i) Evaluate the given integral by applying reduction formula 2

$$I = \int_0^{\pi/2} \cos^6 x \, dx$$

(ii) Define Beta and Gamma function. 2

(b) Prove that : 8

$$\beta\left(m, \tfrac{1}{2}\right) = 2^{2m-1}\beta(m, m)$$

(c) Apply the concept of line integral and evaluate area common to the parabola $x^2 = 8y$ and straight line $x - 2y + 8 = 0$. 8

(d) Apply the formula of volume of revolution to evaluate the volume generated by revolution of circle $x^2 + y^2 = a^2$ about x axis. 8

Unit-II

2. (a) (i) State Rolle's theorem. 2

(ii) Evaluate : 2

$$\lim_{x \to 0} \frac{\log_e x}{\cot x}$$

(b) Apply Taylor's series expansion to find expansion of $\log_e x$ in in powers of $(x-1)$. Also evaluate $\log_e(1 \cdot 1)$ correct upto 4 decimal places. 8

(c) Derive the Maclaurins series for $f(x) = e^{\sin x}$, upto the term containing x^4. 8

(d) Apply the concepts of maxima and minima and find out the maximum and minimum values of $f(x) = 3x^4 - 2x^3 - 6x^2 + 6x + 1$ in the interval (0, 2). 8

Unit-III

3. (a) (i) Apply D'Alembert's ratio test and verify whether the series is convergent or not, whose n^{th} term is $\frac{n^2}{2^n}$.

(ii) Define Dirichlet's conditions for Fourier series. 2

(b) Apply Cauchy's root test and verify whether the given series is convergent or divergent? 8

$$\frac{1}{2} + \frac{2}{3}x + \left(\frac{3}{4}\right)^2 x^2 + \left(\frac{4}{5}\right)^3 x^3 + \ldots\infty \quad (x > 0)$$

(c) Obtain Fourier series expansion of $f(x) = x - x^2$, $-\pi \le x \le \pi$. Also show that 8

$$\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\ldots\ldots=\frac{\pi^2}{12}$$

(d) Expand $f(x)=x\sin x$, in $(0, \pi)$ as half range cosine series and also show that 8

$$\frac{1}{1\cdot 3}-\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}-\ldots\ldots\infty=\frac{\pi-2}{4}$$

Unit-IV

4. (a) (i) Compute the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f(x, y)=x^y$. 2

(ii) Evaluate $\nabla\cdot R$ and $\nabla\times R$, where $R=xi+yj+zk$. 2

(b) Show that :

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u(x, y, z)=\frac{-9}{(x+y+z)^2}$$

where

$$u(x, y, z) = \log_e \left(x^3 + y^3 + z^3 - 3xyz\right)$$ 8

(c) A rectangular box open at the top is to have volume of 32 cubic ft. Find the dimensions of the box requiring least material for its construction. 8

(d) Compute a and b for which the surfaces $ax^2 - byz = (a + 2)x$ and $4x^2y + z^3 = 4$, cut orthogonally at $(1, -1, 2)$. 8

Unit-V

5. (a) (i) Define Rank of a matrix. 2

(ii) Find a modal matrix P, which diagonalize the matrix

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

It is given that eigen values of A are 2 and 5. 2

(b) Evaluate λ and μ for which the given system of equations

$$x+y+z=6$$

$$x+2y+5z=10$$

$$2x+3y+\lambda z=\mu$$

having unique solution, no solution and infinite many solutions. Also find particular solution for $\lambda=2$ and $\mu=8$. 8

(c) Compute the eigen values and corresponding eigen vectors for $A=\begin{bmatrix} 10 & -2 & -5 \\ -2 & 2 & 3 \\ -5 & 3 & 5 \end{bmatrix}$. 8

(d) Verify Cayley-Hamilton theorem for the matrix

$$A=\begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$$

and also compute A^{-1}. 8